

Clinical Risk Assessment Report: Patient 90

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.53x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.6%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.13 [Avg (63th %ile)] (-1.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 63.0 [Avg (47th %ile)] (-1.0%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.29 [Top 10%] (-1.0%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -1.0 [Bottom 17%] (-0.8%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.57 [Avg (28th %ile)] (-0.7%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.08 [Avg (67th %ile)] (-0.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: -0.75 [Bottom 21%] (-0.3%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.81 [Avg (46th %ile)] (-0.1%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Machine Learning Feature Attribution (SHAP Decision Path)

